

Automatic Detection and Localization of Endometriosis in Ultrasound Videos

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Abstract

Endometriosis is a chronic condition affecting approximately 10 percent [1] of reproductive-age female individuals, often leading to severe pain and infertility. Despite its prevalence, diagnosing endometriosis remains challenging, with delays ranging from 4 to 11 years due to the reliance on invasive laparoscopic procedures. This thesis relies on an endometriosis detection pipeline created in a previous semester project by Théo Ducre and focuses on fine-tuning the RAFT optical flow network to pelvic ultrasound videos. Specifically, the work aims to improve the capability of the RAFT model to accurately estimate optical flow from ultrasound data, thereby enhancing the identification and localization of endometriosis. Multiple preprocessing techniques, such as speckle noise reduction and contrast enhancement were systematically evaluated to optimize input quality for the RAFT neural network. The model was fine-tuned using different loss functions, including Mean Absolute Error, Structural Similarity and Forward Backward Consistency Loss. Synthetic ultrasound data was also generated using parametric shapes with speckle noise and elastic deformation of real ultrasound images to validate the model's performance in a controlled environment. Experimental results show that SSIM loss yields the best results and that some preprocessing techniques, like SRAD, work better than others. This thesis highlights the critical impact of fine-tuning parameters and preprocessing methods on model performance, laying a foundation for future research aimed at developing non-invasive diagnostic tools for endometriosis using advanced deep-learning techniques.

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Abbreviations

RAFT	Recurrent All-Pairs Field Transforms
US	Ultrasound
GRU	Gated Recurrent Unit
SMURF	Self-Teaching Multi-Frame Unsupervised RAFT with Full-Image Warping
SAM	Segment Anything Model
UnSAM	Unsupervised Segment Anything Model
VolRAFT	Volumetric RAFT
SRAD	Speckle Reducing Anisotropic Diffusion
HE	Histogram Equalization
AHE	Adaptive Histogram Equalization
CLAHE	Contrast-Limited Adaptive Histogram Equalization
CDF	Cumulative Distribution Function
FBCL	Forward-Backward Consistency Loss
HSV	Hue Saturation Value
SSIM	Structural Similarity
MAE	Mean Average Error

1 Introduction¹

Endometriosis is a medical condition that affects approximately 10 percent of women of reproductive age [1], and is well known to be difficult to diagnose using ultrasound (US). The gold standard for confirming the diagnosis remains laparoscopic surgery. This procedure is invasive and is usually delayed 4 to 11 years [2] due to the complexity and subtlety of the symptoms. Women with endometriosis develop tissue that closely resembles endometrial tissue outside the uterus, usually in other reproductive organs inside the pelvis or in the abdominal cavity. Each month, this misplaced tissue responds to the body’s hormonal cycle, undergoing the same build-up and breakdown as the endometrial lining inside the uterus. However, because ectopic tissue does not have an outlet for the blood and debris it generates, the process leads to internal bleeding that penetrates the cavity. This process in turn causes inflammation, swelling, and irritation of surrounding tissues, which eventually result in scar tissue formation and adhesions. Consequently, women with endometriosis often experience chronic pain, excessive menstrual bleeding, and unexplained infertility [3]. Laparoscopy remains the most reliable method for both diagnosing and treating endometriosis, as it allows direct visualization and potential removal of lesions. In contrast, pelvic ultrasound is a noninvasive diagnostic tool that uses sound waves to produce images of internal reproductive organs, including the uterus, cervix, vagina, fallopian tubes, and ovaries. Although ultrasound offers a safer and more accessible alternative to laparoscopy, its sensitivity in detecting endometriosis is limited, especially in early or subtle cases [4].

The basis of this thesis is a pipeline developed in a previous semester project. This pipeline uses the basic RAFT model [5] trained on Things/Sintel 2.2. This work was carried out in collaboration with SCANVIO Medical, and the objective is to improve the current capabilities of the RAFT model to extract the optical flow of ultrasound videos of the pelvis region.

1.1 Background and Motivation

1.1.1 Understanding Endometriosis

Endometriosis occurs when tissue similar to the uterine lining (the endometrium) implants outside the uterus, most commonly throughout the pelvic cavity [6]. Each month, this ectopic tissue continues to thicken, break down, and bleed in response to the menstrual cycle, but unlike the uterine lining, it cannot be expelled through the vagina. As a result, trapped blood and debris cause inflammation, scar formation, and adhesive bands that can bind pelvic tissues and organs, leading to further complications. Affected individuals often experience chronic pelvic pain, pain during intercourse, painful bowel movements or urination, heavy menstrual bleeding, and even infertility [3]. They may also experience fatigue, nausea, diarrhea, constipation, and bloating, with most symptoms worsening around the time of their period.

¹Parts of this introduction are inspired by Jesse Schweizer and Théo Ducre.

1.1.2 Challenges in Diagnosing Endometriosis

Endometriosis, even though it affects a comparable number of people and causes similar profound health consequences, has traditionally attracted much less research funding than other conditions of equivalent prevalence and severity, as Kirk et al. [7] have demonstrated. In the United States during 2022, for example, diabetes research received approximately fifteen times more funding per female patient than endometriosis, while Crohn’s disease, another chronic inflammatory disease, obtained more than sixty-five times the budget per patient.

One of the most troubling consequences of this underinvestment is the prolonged interval between the first of symptoms and the point of definitive diagnosis, which can range from 4 to 11 years [2]. Several factors contribute to this diagnostic delay [8]. First, the ‘gold standard’ to confirm endometriosis remains laparoscopic inspection: an invasive surgical procedure requiring general anesthesia. Due to its invasive nature, laparoscopy is rarely performed until years after symptoms first appear.

Second, many symptoms, such as severe menstrual cramps, pelvic discomfort, and gastrointestinal distress, have long been dismissed as ‘normal’ aspects of menstruation. This normalization can lead both patients and medical professionals to downplay warning signs, delaying further investigation. Third, the heterogeneous presentation of endometriosis means its pain and dysfunction frequently overlap with other disorders, and misdiagnoses are common; in some cases, endometriosis is mistakenly treated as Crohn’s disease or irritable bowel syndrome, prolonging the path to correct treatment [9].

Although pelvic ultrasound represents a promising, non-surgical and cost-effective screening tool, its precision depends on the specialized expertise of the sonographer in recognizing subtle markers of endometriotic lesions [10]. To date, no noninvasive diagnostic modality has demonstrated sufficient sensitivity and specificity to replace laparoscopy to identify deep-infiltrating endometriosis. As Zhang et al. [11] have reported, existing imaging techniques and biomarker assays still do not replace surgical evaluation as the definitive means of diagnosis.

Together, chronic underfunding of endometriosis research, coupled with diagnostic challenges and symptom normalization, perpetuates a cycle in which women and those assigned women at birth endure unnecessary suffering and delays in effective care. Addressing these issues will require not only increased financial investment in endometriosis-specific studies but also greater awareness among clinicians and patients, more robust development of noninvasive diagnostic methods, and standardized training for imaging specialists.

1.2 Research Objectives and Contributions

Given Théo Ducre’s existing pipeline, the first goal of this project is to find out if the RAFT model is capable of learning flow fields from ultrasound images. This is done by overfitting the model on the data. Then, another goal is to use the existing RAFT model and fine-tune it

on ultrasound videos to extract the flow. This work shows what is achievable and what is not possible and which parameters to use.

1.3 Outline of the thesis

The introduction gives an overview of the disease, motivating the need for noninvasive diagnostic tools, and states the research objectives. The related work section reviews the existing approaches and provides a special overview of the RAFT model used in this project. The method section describes in detail the data used, the setup, and the preprocessing. It then goes into more detail about how and why the model was overfitted on the data and describes in detail the fine-tuning pipeline using RAFT as a backbone. This part also mentions how the results are visualized. The results section presents the findings, the discussion and conclusion section reflects on those findings and critically evaluates future work directions.

2 Related Work

2.1 Existing Methods for Flow Estimation and Endometriosis Detection

Recent years have seen an increase in AI models for endometriosis diagnosis. On the clinical data side, Bendifallah et al. [12] developed machine learning algorithms as a new screening approach based on patient clinical features, achieving high accuracy and suggesting a shift to an AI-driven endometriosis detection. A review by Wei et al. [13] shows both the promises and the pitfalls of AI in endometriosis, emphasizing that while various models show potential, robust validation and data heterogeneity are essential.

On the medical imaging side, researchers are exploring the use of AI to interpret ultrasound and other imaging modalities for the diagnosis of endometriosis. Several studies have been conducted on ultrasound images because it is an easy and accessible tool to use. Balica et al. [14] optimized five of the most widely used deep learning methods on classifying endometriosis from ultrasound images. Their results suggest that deep learning could indeed analyze patient data and classify endometriosis with high accuracy.

In the paper by Tian et al. [15], the authors propose a coarse-to-fine-based, end-to-end method to estimate optical flow in an unsupervised manner. The architecture uses a feature pyramid to extract features at different resolutions and an optical estimator, then estimates the forward and backward flow. This method provides a good approach but is lacking in publicly available code and was also only ever used in a liver Ultrasound video. SMURF [16] is a RAFT [5] based architecture that takes a three-image input and provides flow forward and backward predictions accounting for occlusions. This method is an improvement over the original RAFT model architecture, but is also slower, since it uses two passes in the RAFT model and builds a frame model for flow prediction. In their paper, Lee et al. [17] use multiple frames to reconstruct organs from ultrasound videos. Although this doesn't extract movement, its use was briefly considered. UnSAMFlow [18] is a promising approach that uses segmentation to extract the flow. UnSAMFlow combines unsupervised optical flow with SAM [19] to identify flow in ultrasound videos. Although this approach is also very interesting, it is not explored further in this work. The paper by Abhimanyu et al. [20] presents the URAFT model for deformable registration of ultrasound images. They train the model, which is composed of RAFT and a spatial transformer network (STN) in an unsupervised way to predict flows. This model is useful for compensating different motions like respiration or heartbeat in the images. Finally VolRAFT [21] is the RAFT architecture applied to 3D volumes. This is useful to capture the subtleties of 3D motions of organs in videos, but is not applicable to this project.

2.2 Existing Pipeline

This thesis builds on a semester project that provides the whole pipeline. First a segmentation model provided by Scanvio is used to create a segmentation map of the uterus region. Points

are sampled uniformly at random along the uterus boundary. The framework then uses the optical flow provided by RAFT to extract the flow. This flow is normalized and compared to the previous frame, where small flows are filtered out. A Laplacian filter with a threshold is used to detect strong variations at each point. A strong variation means that the region is sliding, hence it is most likely a healthy region. The total number of healthy points in each frame is then counted, and the frame is classified using a threshold on the count of healthy points.

2.3 RAFT Architecture

The RAFT, Recurrent All-Pairs Field Transforms architecture, was proposed by Teed and Deng [22] in 2020 and presents a novel neural network architecture to extract optical flow from images. The network is conceptually divided into a feature extraction stage, a full all-pair correlation stage, an iterative update operator that refines a single high-resolution flow field, and a learned upsampler that raises the $1/8$ -scale predictions back to full resolution. Below each component is reviewed and design decisions that allow RAFT to achieve state-of-the-art accuracy with only ≈ 5.3 M parameters are highlighted.

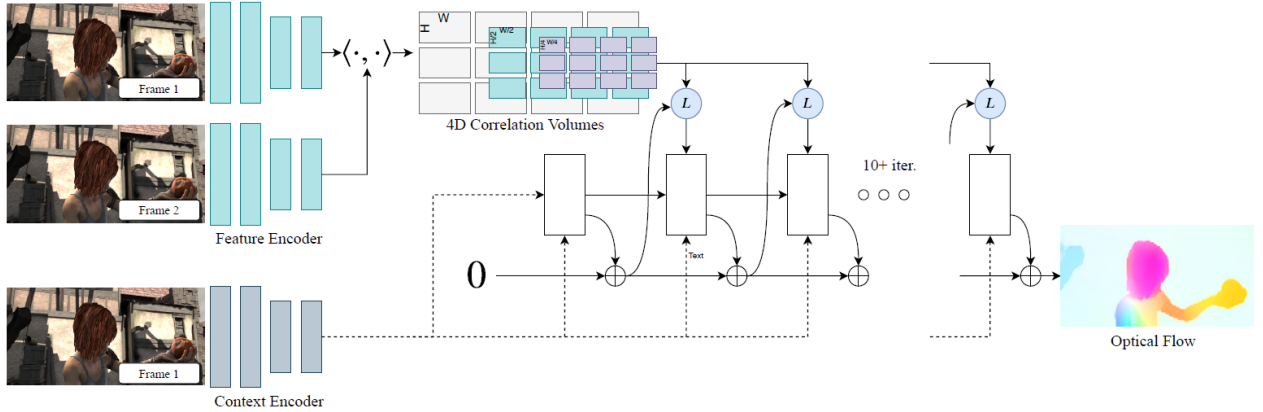


Figure 1: Architecture of the RAFT (Recurrent All-Pairs Field Transforms) model. This diagram shows feature extractor, correlation volumes, recurrent update operator.

2.3.1 Overview & Design

Traditional variational and discrete optimisation methods alternate between computing a data-term and a smoothness prior, often within a coarse-to-fine pyramid. RAFT mimics that loop in a learnable way.

1. All-pairs matching is pre-computed once via a 4-D correlation volume that captures similarity scores between every pixel in the first image and every pixel in the second; this replaces repeated warping or cost-volume building at run time.
2. A recurrent GRU-style block with then plays the role of an optimiser, retrieving local slices of the correlation pyramid and predicting an incremental flow update at each iteration. Because weights are shared, the block can run for an arbitrary number of iterations

(the authors report up to 200 without divergence), analogous to unrolling many steps of gradient descent.

3. A convex-combination upsampling layer outputs dense flow at full resolution while preserving sharp motion boundaries.

2.3.2 Feature and Context Encoder

Let the input RGB frames be $I_1, I_2 \in \mathbb{R}^{H \times W \times 3}$. A shared convolutional feature encoder g_θ produces feature maps

$$F_1 = g_\theta(I_1), \quad F_2 = g_\theta(I_2), \quad F_i \in \mathbb{R}^{256 \times H/8 \times W/8}$$

Those feature maps transform the input image into dense feature vectors. The encoder works as follows: First the image is passed through a 7x7 convolution-norm-activation block mapping the input image of the form $C \times H \times W$ to $64 \times H/2 \times W/2$. Then it passes through 3 Residual units, the first one with stride 1 and the two others with stride 2, the image outputted by the residual blocks is of shape $256 \times H/8 \times W/8$. Finally a last convolutional layer gives the image its final shape of $256 \times H/8 \times W/8$. A second context encoder h_θ processes only I_1 . It is architecturally identical to the feature encoder.

Its output is split into a hidden state initializer for the GRU and static per-pixel context features concatenated to the GRU input at every iteration, injecting image semantics into the optimisation loop.

2.3.3 4-D Correlation Volume and Pyramid

A correlation volume is computed between all the pairs, given the extracted features $g_\theta(I_1)$ and $g_\theta(I_2)$. The correlation $\mathbf{C}$ is given by:

$$C(g_\theta(I_1), g_\theta(I_2)) \in \mathbb{R}^{H \times W \times H \times W}, \quad C_{ijkl} = \sum_h g_\theta(I_1)_{ijh} g_\theta(I_2)_{klh}.$$

This is computed by a single matrix multiplication. To balance memory with large-motion capture, the last two axes are pooled with kernel sizes 1,2,4,8 to form a four level correlation pyramid C^1, C^2, C^3, C^4 . Hence the volume C^k has dimension $H \times W \times H/2^k \times W/2^k$. The first two axes are never down-sampled, preserving high-frequency detail and allowing small objects to be tracked even when large displacements are present.

2.3.4 Recurrent GRU-based Update

This flow is used to retrieve the correlation features from the correlation pyramid $\mathbf{C}$. The correlation features and the estimated flow f^t at iteration t then go through 2 convolutional layers. A combination x_t of the correlation, the predicted flow and a hidden state is then used

as input for the GRU block. The GRU update operator is a gated activation unit with fully connected layers replaced with convolutions, which outputs a hidden state. This hidden state is passed through two convolutional layers to predict the flow update Δf . The output flow is still at 1/8 resolution and is upsampled before being returned. The upsampling module takes for each high resolution flow field a convex combination of its nine coarse resolution neighbors using weights predicted by the network.

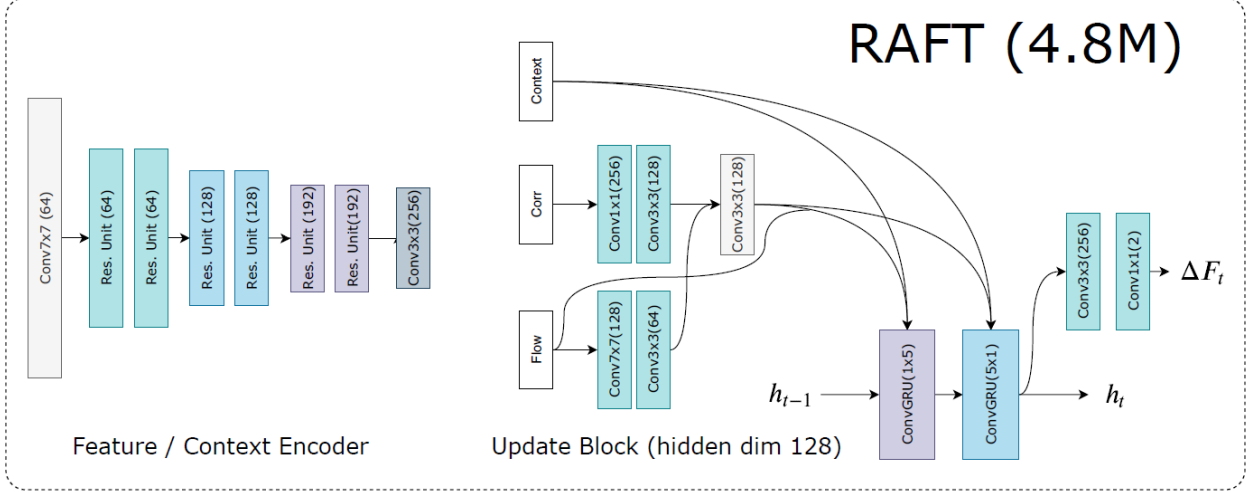


Figure 2: This diagram shows the details of the feature encoder and the updated block.

2.3.5 Limitations

The memory is a big issue since pixel count scales quadratically with image size. The authors found a way to precompute correlations to speed up the computation process. Another alternative they mention is to compute each correlation volume on demand. This helps soothe the problem, but not mitigate it completely, as the image size is still a problem.

3 Method

Here is a brief overview of the steps in the project.

3.1 Data & Experimental setup

The data used consists of ultrasound videos provided by Scanvio. The dataset is filtered to only contain videos that show sliding. The sliding sign is an ultrasound technique where one gently pushes the cervix region with a trans-vaginal probe in order to find out if the anterior rectum glides freely against the posterior aspect of the cervix and posterior vaginal wall [23]. A smooth sliding is detected, and the sliding sign is considered positive. In total, there are 3623 videos with this sliding sign, ranging from 19 to 1695 frames in length, with an average length of 220 frames. Each frame has a resolution of 1528x784. Additionally, a few handpicked videos help visualize the results.

3.2 Preprocessing

Before feeding ultrasound frames into the RAFT network, a standardized sequence of preprocessing steps to reduce noise, normalize intensity distributions, and enhance structure visibility is applied.

3.2.1 Intensity Normalization

Each input pair I_t, I_{t+1} is linearly mapped to the $[-1, 1]$ range using the formula:

$$\hat{I} = \frac{I - \mu}{\sigma}$$

This process ensures that $\hat{I} \in [-1, 1]$, normalizes raw 8-bit intensities $I \in [0, 1]$, and maintains stable training dynamics under RAFT’s original encoder weights.

3.2.2 Speckle Noise Reduction

Ultrasound images are corrupted by multiplicative speckle noise, which degrades flow estimation if left untreated. For methods that require additive noise instead of multiplicative noise, a logarithmic transform that converts multiplicative noise into additive noise is used, by first taking the logarithm of the image, then applying an additive transformation, and then using an exponential to convert it back to its original domain. The three denoising approaches are:

1. **Speckle Reducing Anisotropic Diffusion (SRAD)** This method was first introduced by Yu and Acton [24]. SRAD solves the partial differential equation in an iterative manner

$$\begin{cases} \frac{\partial I(x, y, t)}{\partial t} = \text{div}(c(q) \nabla I(x, y, t)) \\ I(x, y, 0) = I_0(x, y), \quad \left. \frac{\partial I(x, y, t)}{\partial \vec{n}} \right|_{\partial \Omega} = 0 \end{cases}$$

SRAD not only keeps edges intact but also makes them clearer by blocking smoothing across them and letting it happen along each side. It adjusts itself without using fixed cut-offs, so it works well in both flat areas and around edges or small details. This method, however, is computationally intensive as it requires iteratively solving PDEs.

2. Wavelet (Haar) Denoising

This method from Qiang [25] decomposes I into Haar wavelet coefficients W . The Haar transform is defined as:

$$H_m = [h_m(\frac{2p-1}{2n})]_{p=1, \dots, m}$$

where

$$h_m(x) = [h_0(x), h_1(x), \dots, h_{m-1}(x)]^T.$$

$$h_0(x) = \begin{cases} 1, & x \in [0, 1], \\ 0, & \text{otherwise}, \end{cases} \quad h_i(x) = \begin{cases} 1, & x \in [\frac{k}{n}, \frac{k+\frac{1}{2}}{n}), \\ -1, & x \in [\frac{k+\frac{1}{2}}{n}, \frac{k+1}{n}), \\ 0, & \text{otherwise}. \end{cases}$$

and where $i \in 0, 1, 2, \dots, 2^{j+1} - 1$

$n = 2 \times j, j \in \mathbb{N}$

$i + n = k$

Hence, the matrix C is constructed, where $C = H_n F H_n^T$. C is split into four blocks, and a threshold is applied. The inverse Haar transform can then be used to reconstruct the denoised image.

3. Median Blur

A non-linear, edge-preserving filter replaces each pixel by the median $k \times k$ neighborhood. This method is good at removing salt-and-pepper type noise and is also very fast, but it blurs fine detail and is not tailored for multiplicative noise

3.2.3 Contrast Enhancement

Global contrast differences across scanning depths can impede optic-flow learning. Two histogram-based methods are used:

1. Histogram Equalization (HE)

HE remaps intensities to flatten the global histogram:

$$\hat{I} = T(I) = \text{CDF}_I(I),$$

Where $\text{CDF}_I(I)$ is the empirical cumulative distribution.

This is a simple method to increase overall contrast but amplifies noise uniformly as speckles become more pronounced.

Adaptive Histogram Equalization (AHE) differs from HE by computing several histograms to distribute the lightness values of the images. This helps improve local contrast and edges in the image.

2. Contrast-Limited Adaptive Histogram Equalization (CLAHE)

Introduced by Pizer et al. [26], this method uses AHE but clips the histogram at a predefined value to limit the slope of the CDF. This avoids big, sharp slopes for near-constant regions where most pixels have the same values.

3.2.4 Other Methods

Log Transform converts multiplicative speckle noise into an approximation of additive noise. It compresses the dynamic range, making low-intensity structures become more visible. This methods can however over-compress shadows and boundaries, reducing contrast where it might be needed.

Masking out parts of the video or cropping the videos was also considered and tested out to some extent, but this approach was inevitably dropped as the number of videos available was deemed sufficient, and there was no need to enhance the dataset with such artificial data.

3.3 Losses

To train the flow estimation network, several losses are used and combined. The losses target pixel-wise accuracy, perceptual similarity, and geometric consistency. Let $\mathbf{F}_f$ be the forward flow predicted from frame I_t to I_{t+1} and $\mathbf{F}_b$ be the backward flow from I_{t+1} back to I_t . Let $\hat{I}_{t+1} = W(I_t, \mathbf{F}_f)$ denote the frame I_t warped forward under $\mathbf{F}_f$.

3.3.1 Mean absolute Error (L1)

The L1 loss penalizes the pixel-wise difference between the warped image and the target:

$$\mathcal{L}_{L1} = \frac{1}{N} \sum_{n=1}^N |\hat{I}_{t+1}(n) - I_{t+1}(n)|$$

Where N is the total number of pixels and n indexes each pixel location. This loss encourages overall brightness consistency and is robust to outliers compared to an L2 penalty.

3.3.2 Structural Similarity (SSIM)

This loss measures perceptual quality over small patches. The SSIM is given by:

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)}$$

where:

μ_x is the pixel sample mean of x

μ_y is the pixel sample mean of y

σ_x^2 is the sample variance of x

σ_y^2 is the sample variance of y

σ_{xy} is the sample covariance of x and y

$c_1 = (k_1L)^2$ and $c_2 = (k_2L)^2$ are two variables to stabilize the division with a weak denominator

L is the dynamic range of the pixel values

By default, $k_1 = 0.01$ and $k_2 = 0.03$

3.3.3 Forward-Backward Consistency Loss (FBCL)

FBCL is used to force a certain consistency in the flow. Instead of just calculating the flow from the first to the second image with the model, the same images are reused to also calculate the flow from the second to the first image. The consistency says that for every pixel x in the first image I_t , if you follow the forward flow $\mathbf{F}_f(x)$ to its corresponding location in the second image and then apply the backward flow $\mathbf{F}_b$ at that new position, you should return to the original pixel:

$$\mathbf{F}_b(x + \mathbf{F}_f(x)) \approx -\mathbf{F}_f(x) \quad (1)$$

$$\mathbf{F}_b(x + \mathbf{F}_f(x)) + \mathbf{F}_f(x) \approx 0 \quad (2)$$

The deviation is penalized:

$$\mathcal{L}_{\text{FB}} = \frac{1}{N} \sum_x \|\mathbf{F}_b(x + \mathbf{F}_f(x)) + \mathbf{F}_f(x)\|_1$$

3.4 Synthetic flow generation

To validate the implementation and establish a controlled sandbox for experimentation, RAFT is first trained and evaluated on synthetically generated ultrasound-like sequences. By verifying that the network can recover known fields in a toy setting, confidence is gained before moving

to real-world data. Additionally, synthetic tests help debug training dynamics and overfitting strategies. Two complementary generation strategies were explored.

3.4.1 Approach 1: Parametric Shapes + Speckle Noise

Overview

Video sequences of simple white shapes (circles or rectangles) moving against a dark background sprinkled with speckle like noise are generated. Two background regions, separated by a smooth polynomial boundary, flow with distinct constant velocity vectors. This construction yields ground truth flow fields that vary spatially in a controlled manner.

The key steps are:

1. Polynomial region mask

First, four control points (x_i, y_i) are sampled along the top and bottom edges. A cubic polynomial $y = p(x)$ is then fitted to these points. Finally a soft mask M is created where

$$M(x, y) = \sigma(p(x) - y; w) \text{ with } \sigma(z; w) = \frac{1}{1 + e^{-z/w}} \text{ where } w \text{ controls transition smoothness.}$$

2. Background speckle

To create background speckle noise, two binary noise patterns are used, dilated with an elliptical kernel and blurred with a Gaussian filter. At each frame, these patterns are sampled by offsetting their sampling grid according to two constant flow vectors v_1, v_2 , then blended using M .

3. Animated shapes

N shapes, circles or rectangles, are initialized with random size, position and velocity. On each frame, those shapes are updated in a way that they bounce on the edges of the canvas, and the shapes are also blended on top of the background.

4. Ground-truth flow

For each pixel (x, y) the truth flow is

$$\mathbf{F}(x, y) = M(x, y) \mathbf{v}_1 + (1 - M(x, y)) \mathbf{v}_2$$

An advantage of this synthetic flow generation is that the flow is perfectly known, it has clear edges for pixel-wise losses and it can be generated pretty fast. However, it does not capture real tissue texture and movement.

3.4.2 Approach 2: Elastic Deformation of Real Ultrasound

Overview

To mimic organ motion and speckle noise more faithfully, elastic (B-spline) deformations to actual pelvic ultrasound frames is applied. This produces non-rigid warping and realistic intensity variations.

Here are the key steps

1. **B-spline control grid**

First, a coarse lattice of control points is defined, then each control point is perturbed by a random displacement drawn uniformly from $[-d_{max}, d_{max}]$ in each axis, while fixing the borders to zero displacement.

2. **Resampling**

SimpleITK's image filter is then used with linear interpolation to warp the image under the B-spline transform and compute the corresponding displacement field via their field displacement filter.

3. **Flow visualization**

The 2D flow vectors (u, v) are converted to HSV color encoding, where

$$Hue = \arctan2(v, u)$$

$$Value = \sqrt{u^2 + v^2} \text{ normalized to } [0,1]$$

4. **Sequence generation**

Random deformations are iteratively applied to the original or the previously deformed frame, yielding a short video with smooth, organ-like motion.

This method captures non-rigid warping and realistic speckle interaction by using real life image textures. However, it is still difficult to render real-life movement of organs through elastic deformations alone, and it also doesn't account for the periodic occlusion of some parts because of 3-D movement.

3.5 Visualizing the Performance

Knowing if the flow is a "good" flow without having a ground truth flow is a tricky question. Instead of trying to visualize the flow directly, the idea was to apply the flow on the first image to see if it resembles the second image. This concept is later referenced as warping. Of course, the flow cannot be perfect since US images vary in intensity and because the 3-D movement of the organs, some zones get occluded periodically. To visually see the difference, one easy way is to view the video instead of individual frames. For this purpose, the original video is put next

to the reconstructed video. At step t_1 , both videos show the original first frame. At step t_n , the original video shows the n -th frame from the original frames, while the reconstructed video shows the warped frame $\hat{I}_n$. To account for the different occlusions and intensity changes, the frame $\hat{I}_n$ is hard reset to the frame from the original video every x steps. The longer both videos stay similar, the better. Meaning that the bigger x is and the smaller the visual difference is, the better. This reveals how well the network can predict pixel displacement in the absence of explicit supervision.

To fully visualize the performance, four panels, assembled in a 2x2 grid, are constructed [3]. The top-left panel shows the reconstructed frame $\hat{I}_n$. The top-right panel always shows the original frame. The bottom left panel shows the RGB-encoded flow map and the bottom-right panel shows a live plot of the training loss.

Together, these visualization and analytic tools provide both intuitive and quantitative insights into the networks performance on challenging real-world ultrasound sequences.

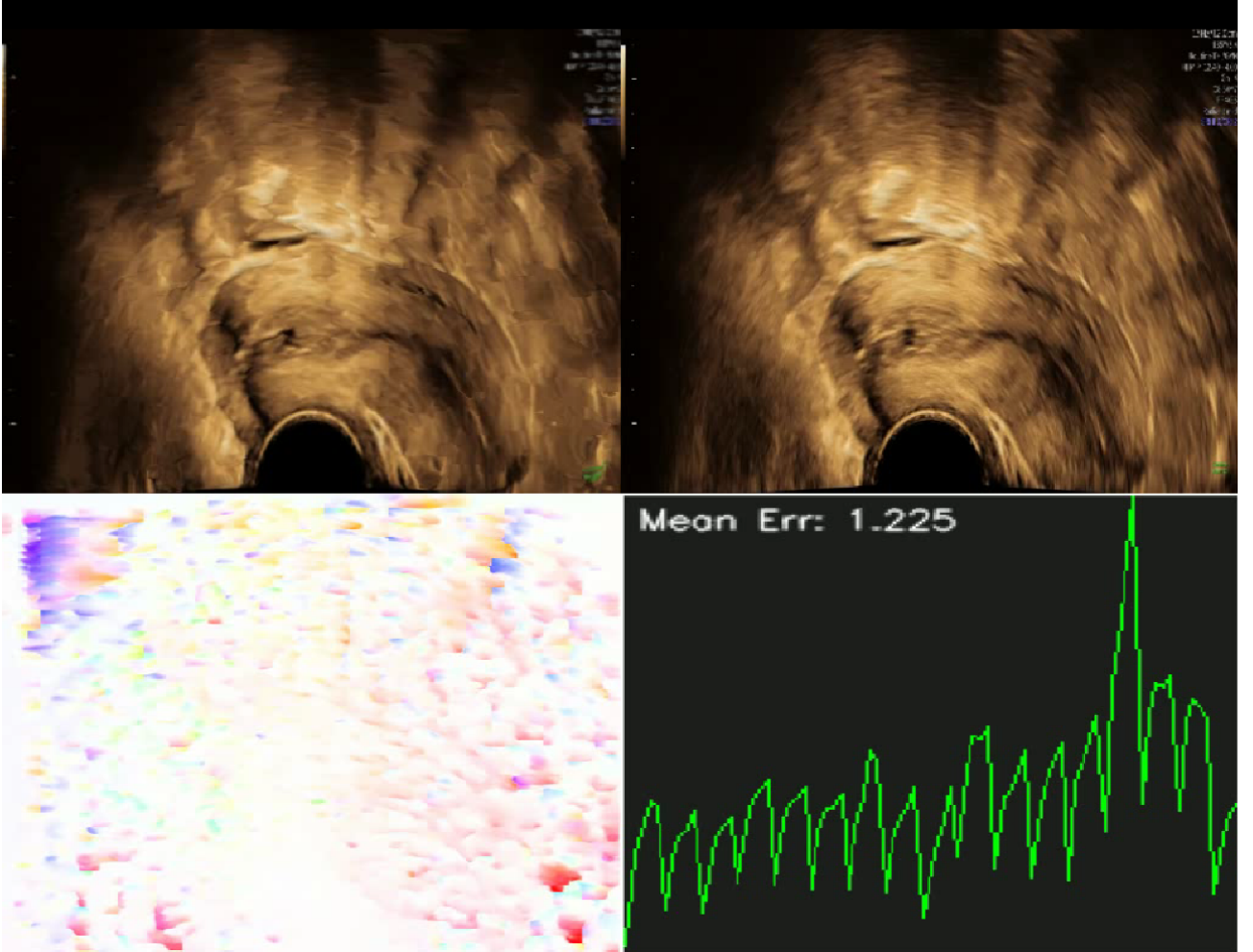


Figure 3: This figure shows an example frame of the mobility panel generated. The top left panel shows the reconstructed picture using the flow, the top right panel shows the original image, the bottom left panel shows the colorized flow field and the bottom right panel shows the loss histogram.

3.6 Overfitting

Overfitting is the last step before fine-tuning the model on the real data. This necessary step helps to verify that the RAFT backbone is capable of memorizing the optical flow patterns in US data. By deliberately training on a small subset of frames, one can observe whether the model can drive the loss to near zero, hence memorizing the flow, indicating it has sufficient capacity to capture the specialized motion. Overfitting was done on a single video consisting of 175 frames. In total 500 epochs were executed per overfit run for the different losses. This step made clear that losses like FBCL have difficulties alone and need to be combined with other losses.

3.7 Finetuning

After confirming that RAFT can memorize optical-flow patterns through overfitting, the next step is to fine-tune the network on the full ultrasound dataset.

First a unified training script that orchestrates data loading, model configuration, loss computation, and optimization in one place is implemented. The RAFT architecture is loaded, optionally from a checkpoint, and is wrapped in PyTorch’s DataParallel to exploit multiple GPUs. Before entering training mode, the feature encoder, a configurable number of context-encoder blocks and all BatchNorm layers can be frozen using a helper method. This allows managing the models size to train only the most important parameters.

Two custom Dataset classes prepare the data. For image-pair training, a custom loader loads adjacent PNG frames, applies the full preprocessing pipeline, resizes the image to a multiple of eight and normalizes it with RAFT’s default transforms. Alternatively, raw video files can also be processed. Optionally, datasets can cap their size to bound memory usage and training time. A standard dataloader then batches these pairs by shuffling during training and pinning memory for throughput.

Within each iteration, the script computes forward and backward flows by calling the model on I_t, I_{t+1} . The chosen loss function (SSIM, MAE, FBCL or a mixture) is chosen. Then warping I_t into $\hat{I}_{t+1}$ is performed via a grid sample helper when needed. This is all optimized using adaptive moment estimation (Adam) with weight decay and an optional cosine or step scheduler and the script also leverages mixed-precision for speed and memory savings. Gradients are clipped to prevent explosion and per-step metrics are logged by a custom logger, which also saves the model checkpoints every two epochs and writes a running loss summary to disk.

This cohesive implementation, with all hyperparameters exposed via argparse flags such as `-denoise` or `-freeze_cent_blocks` ensures that the fine-tuning experiments are both flexible and reproducible. By encapsulating preprocessing, architecture adjustments, loss selection, and training logistics in one script, a clear mapping is maintained between methodological design and the actual code used, thereby reinforcing the validity and replicability of the results.

3.8 Not Implemented and Dismissed Methods

Not everything could be implemented in time, hence this section discusses some of the non-implemented or left-out methods.

Preprocessing Techniques

A few preprocessing techniques are mentioned in section 3.2. A few other papers, like the paper from Ramakers et al. [27] were taken into account, but due to the already sufficient amount of data, it was decided not to implement the enhancing techniques mentioned in the paper. Another paper by Lizzi and Feleppa [28], talking about different preprocessing methods for US video also gave some useful insights, but was not implemented.

Precision and Model Size

In the results section 4, further parameter combinations could be analyzed. This includes which layers of the RAFT model to freeze or not freeze or what size of model to use. Since RAFT actually provides different model sizes and the ability to use mixed precision, these are some interesting parameters to try out, but were dropped due to time constraints.

4 Results

4.1 Experimental Setup

To assess the quality of the trained models, the different fine-tuned models were tested on a manually selected video. To get a consistent loss metric, it was decided to use an average of the FBCL, SSIM, and the L1 loss. Each loss has its advantages, L1 guarantees a brightness consistency, SSIM ensures that the shapes match each other between images, and FBCL ensures the flow is consistent; a mixture of those losses ensures a strong general metric. In total, a combination of two learning rates 10^{-4} and 10^{-5} with CLAHE, HE, HAAR denoise, Median Blur and SRAD as preprocessing techniques and five different losses were used. Overfitting allowed us to show that FBCL could not be used as a standalone loss, hence it had to be combined with another loss. The used losses are the following:

- **FBCL with SSIM**
- **FBCL with L1**
- **L1**
- **MSE**
- **SSIM**

It was deemed sufficient to train the model for 20 epochs, as increasing the number of epochs just slowly decreased the loss. During fine-tuning, the models are saved at a two-epoch interval. This already makes for 50 different possible combinations without counting the epochs, a big number, hence to avoid having a gigantic amount of visualisation data, epoch 2, 10 and 20 from each combination were chosen. As figure 10 shows, the general trend of all the losses over the epochs goes toward 0. Epoch 2 shows the steepest drop and was therefore also chosen. Epoch 10 and 20 are chosen because they provide additional data, specifying whether the models benefits from more epochs. Figure 4 shows the resulting loss curves of each different run during visualization. The legend is omitted, as the findings will be detailed later. One important observation to make is that there is a drop each five frames. This is due to the fact that as explained in Section 3.5, the original frame is hard reset every 5th frame causing those bumps in the loss.

4.2 Training Losses

The current section gives a brief overview of how model fine-tuning went and how the different losses during training behaved. To avoid having too much data, only 4 models are shown in Figure 5. The general trend shows that the losses are converging.

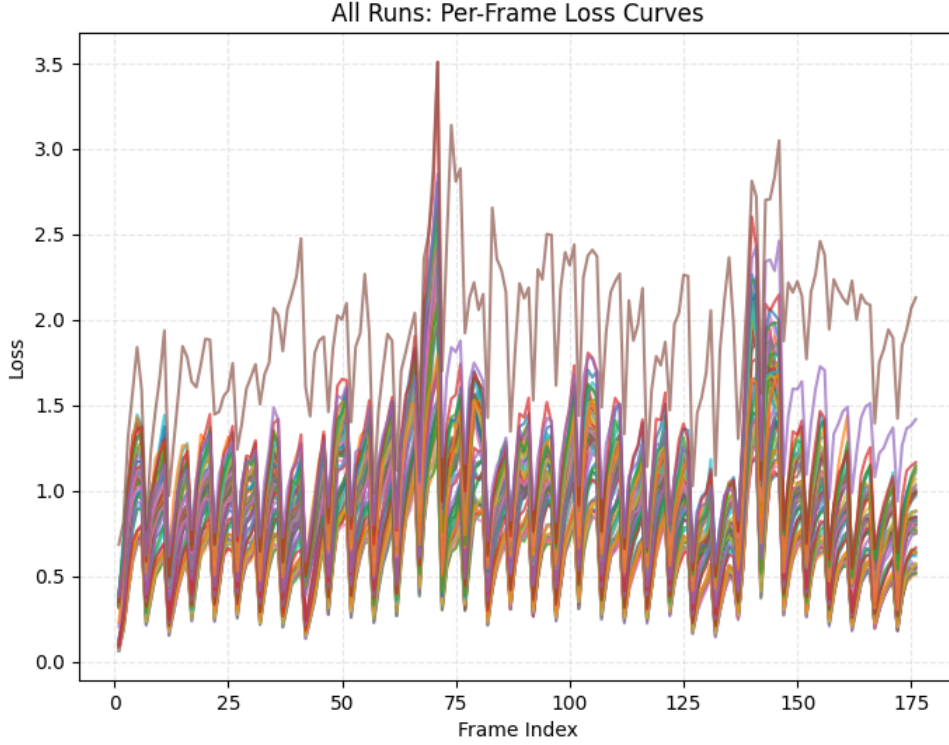


Figure 4: This diagram shows an overview of all the loss curves during visualization

4.3 Assessing the visual performance

Assessing the visual performance is done as explained in section 3.5. The flow using color coding can hence be visualized, each color meaning that the flow field goes in a certain direction. Figure 6 shows a typical flow field. Visualizing the results, however, shows that all flows where the model was fine-tuned using a combination of FBCL loss show signs of a degenerate minimum, meaning that the flow fields go to 0, ie, not showing any displacements at all. Using this information, the next section will present the finding about models trained only using SSIM, MSE and L1 loss and not FBCL loss. Chapter 5 will also discuss potential reasons for a lower score for zero flows and how to avoid it.

4.4 Parameters Impact

This section analyzes the impact of each parameter for fine-tuning mentioned in the experimental setup 4.1.

First the impact of the learning rate was measured. As shown in Figure 7. The losses are grouped by learning rate and the mean is taken for each run. The figure shows that a learning rate of 10^{-5} performs 28% better and reduces the mean loss by ≈ 0.13 . The smaller learning rate helps RAFT avoid overshooting.

As shown in Figure 8, the different preprocessing techniques also play a role in the performance of the model. The lowest and best-performing preprocessing technique is SRAD. This advanced technique although slower than the other performs $\approx 9\%$ better than Histogram Equalization.

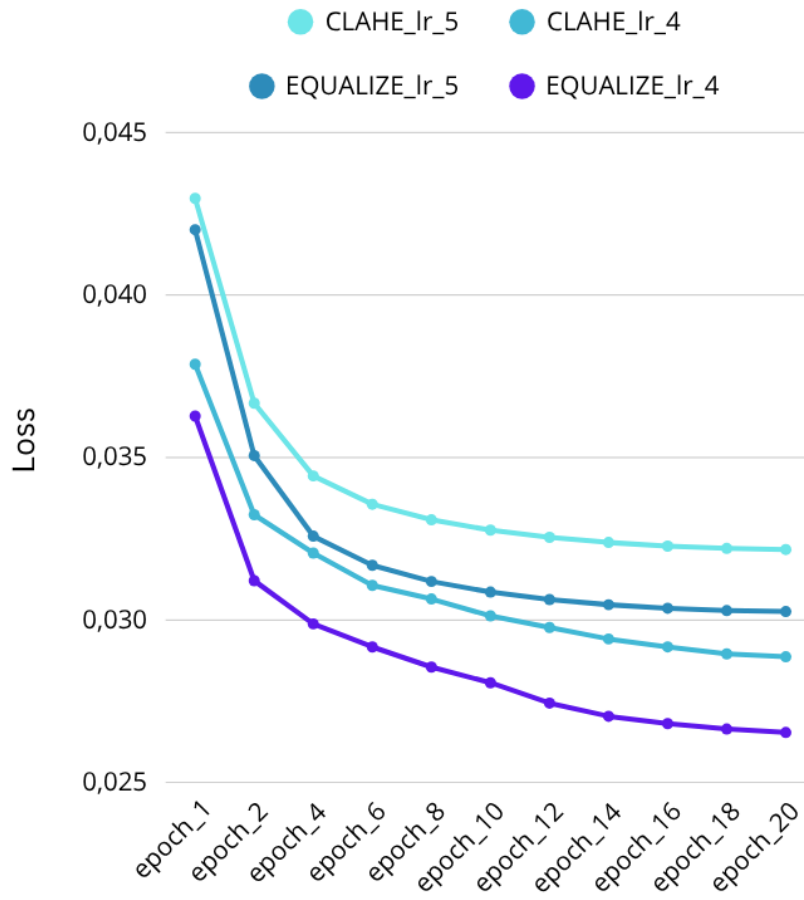


Figure 5: This diagram shows an overview of some of the loss curves during training. The models are handpicked to avoid data overload.



Figure 6: This figure shows what a flow field looks like. The different colors mean different flow directions.

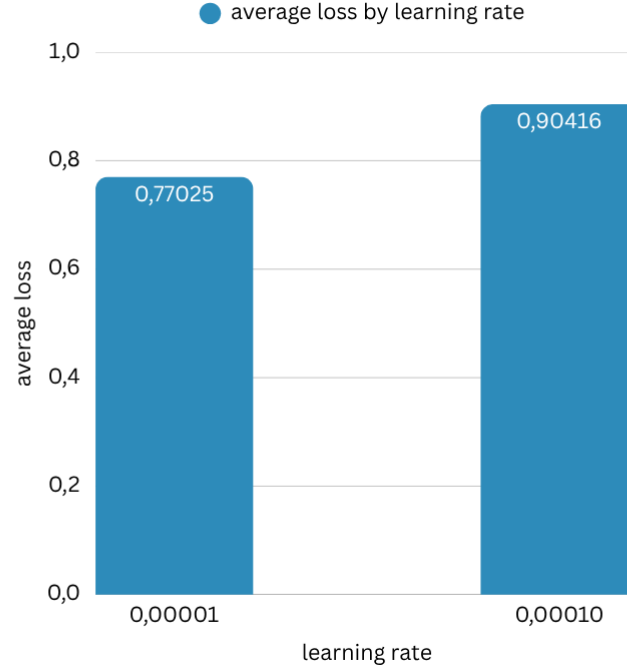


Figure 7: This figure shows the impact of both learning rates 10^{-5} and 10^{-4} on the loss

Haar denoise is also a strong contender and is a nice transition from efficiency to performance. The two worst performing preprocessing techniques, namely CLAHE and Histogram Equilisation perform approximately the same.

The last influencing parameter for fine-tuning is the chosen loss. Figure 9 shows a huge difference between using only the L1 loss and the other losses. SSIM is the strongest contender with L1 performing nearly 50% worse than SSIM.

Figure 10 shows that models trained for only 2 epochs are the best in general and that models trained for 10 epochs benefit from a better performance compared to models trained for 20 epochs.

Figure 11 shows the respective impact of each parameter category. A huge impact is made by the loss function meaning that choosing the right loss is very important for fine-tuning the model.

4.5 Best Performing Combinations

Figure 12 shows the top 10 best-performing models and the top 5 worst-performing ones. The first important detail to note here is that all the top models use SSIM as a loss. This is in accordance with the previous findings shown in Figure 9. Similarly the best model uses SRAD and a learning rate of 10^{-5} . Epoch 2 also dominates the top of the figure's ranking, taking the two first spots.

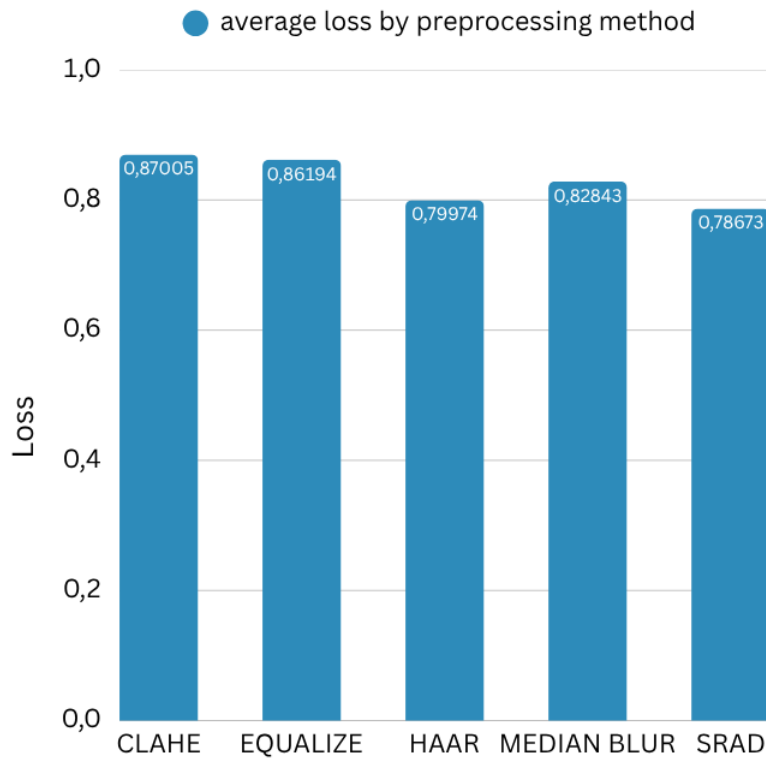


Figure 8: This figure shows the impact of the different preprocessing techniques.

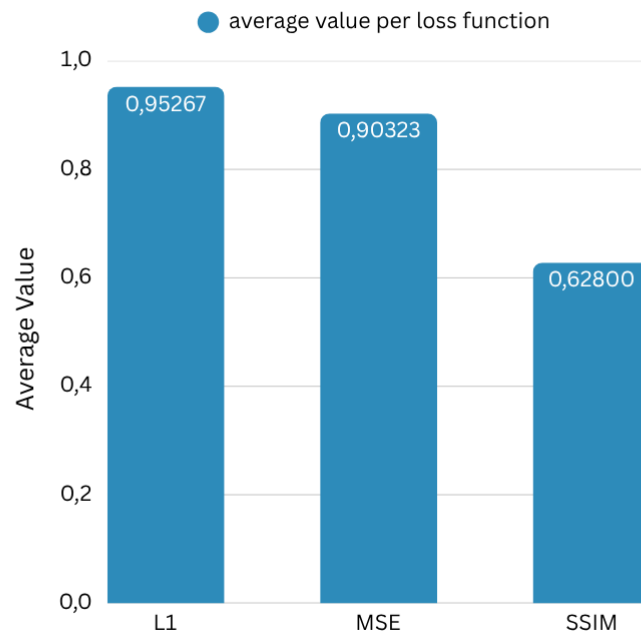


Figure 9: This figure shows the impact of the different losses.

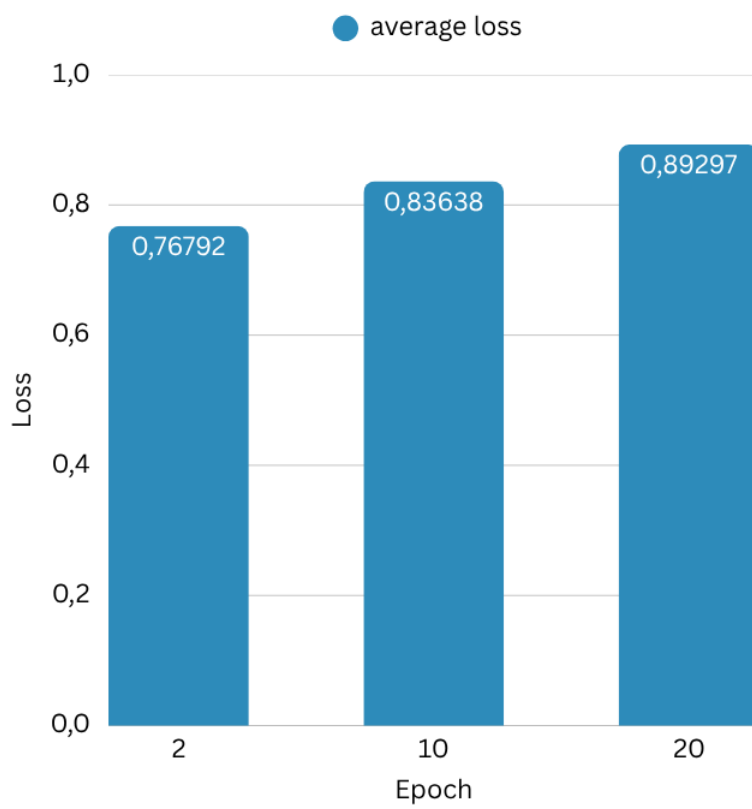


Figure 10: This figure shows the impact of models trained for 2, 10 and 20 epochs.

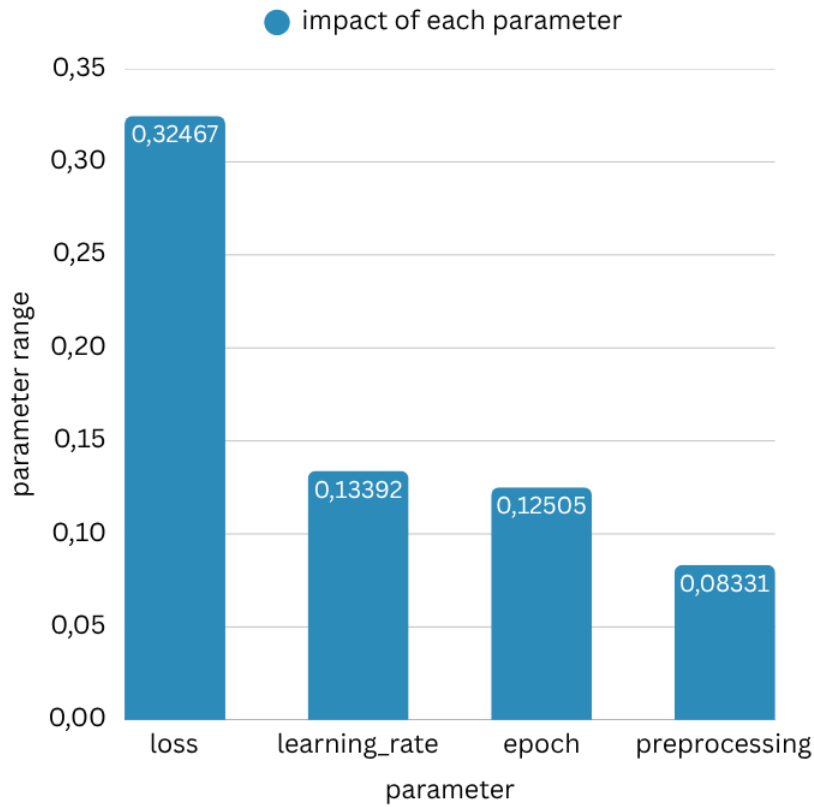


Figure 11: This figure shows the impact of the different hyperparameters.

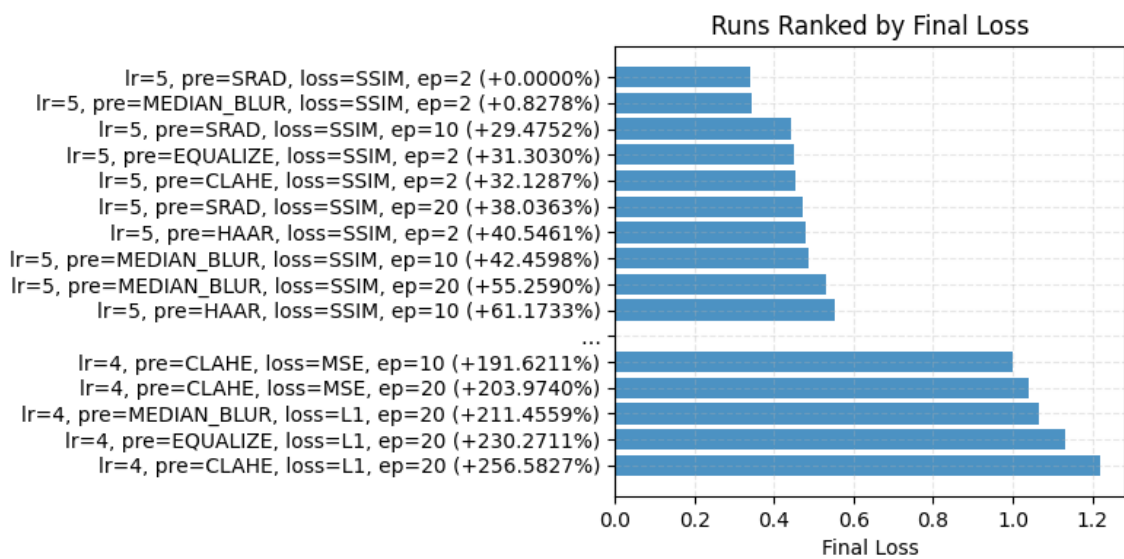


Figure 12: This figure shows the ranking of the best-performing fine-tuned models. The percentage shows how much "worse" the model is compared to the best-performing one

5 Discussion and Conclusion

5.1 summary of key findings

The previous section presents the results that will be discussed here. First when visualizing the performance, it was found out that the flow shows signs of a degenerate minimum going to the 0 flow. This was only for models that used a loss containing the FBCL loss. As seen in overfitting the FBCL loss also went to the 0 flow and that's why it was decided to combine it with other losses. The combination was averaged but further hindsight shows that the FBCL is too powerful compared to the other losses and that a weighted sum of the losses with a much smaller weight for the FBCL loss should have been used. Another insight shows that the model clearly benefits from a learning rate of 10^{-5} instead of 10^{-4} . This smaller learning rate helps the model converge faster and has a big impact on the visualization loss. SRAD shows to be a clear contender for the preprocessing techniques, although slower it does give better scores in general. Another important information is that the models seem to be overfitting at later epochs since the loss increases at epoch 10 and 20 compared to epoch 2. Finally, what the results visually show is that the flows predicted are less coherent, there are more small flows in different directions instead of bigger flows that form the outline of different organs. This however looks visually better, since organs also don't move homogeneously and occlusion is still a persistent issue.

5.2 Limitations and Challenges

Visualizing the flow results was a big part of this project. The lack of ground truth flow data complicated the development of an objective evaluation metric. The RAFT model is still limited by the number of its input parameters since doubling the image size squares the 4D convolution. This leads to increased memory usage for the model and longer processing time of each RAFT iteration. Debugging proved to be particularly time-consuming, ultimately necessitating a complete re-implementation of the codebase to eliminate persistent errors and ensure correct testing. Choosing a good loss is still the biggest challenge as the impact is tremendous on the resulting fine-tuned models. A custom loss, forcing larger flows would be necessary but couldn't be tested due to time reasons.

5.3 Future Research Directions

The current approach falls short in extracting flows that form coherent organ shapes. Instead, as shown earlier, there are a lot of smaller flows that together form a vague form of the different organs in the pelvis region. Since the pipeline then uses a threshold on the uterus boundary, having a lot of smaller flows that more actually represent the visual flow can however hinder the use of a threshold. Alternatively, using the base RAFT model that predicts less accurate results but keeps the flow shapes tighter together might be an easier solution for that pipeline. While some losses have been explored, not all could have been implemented. One interesting

loss to try out would be a loss that penalizes small contradictory flows and encourages bigger, more consistent flows.

The key problem of the fine-tuning part is that there is no ground truth flow available. A solution for this problem would be to create and simulate the motion of 3D organs and use that synthetic flow to fine-tune the RAFT model. 2D simulations of organs wouldn't be sufficient since the ultrasound videos show a 3D motion.

Another interesting approach would be to try out whether RAFT is also able to learn from US videos that don't show the sliding sign. Having a more general model capable of predicting endometriosis independent of what is happening during the recording of the ultrasound would be a great benefit.

Finally, expanded experimentation with RAFT's hyperparameters remains untapped due to time constraints. Systematic parameter sweeps, training on a broader variety of sequences and extending both the number of training epochs and the diversity of input videos by not only using sliding videos could yield substantial gains. These efforts would provide a deeper insight about the models capabilities.

5.4 Conclusion

This work explored the potential of using RAFT neural network architecture to extract accurate optical flow from pelvic ultrasound videos in the context of endometriosis detection. The experimental results demonstrated the critical importance of selecting an appropriate loss function and other hyperparameters, highlighting SSIM as superior to L1 or MSE. This work also demonstrated that, although visual performance is good, concrete benefits for the whole pipeline are not that useful. Future work directions have two main choices: continue fine-tuning RAFT with the right parameters and a loss that forces better flow integrity or explore a new direction not necessarily counting on the flow to diagnose endometriosis. Ultimately, beyond the immediate context of endometriosis, this thesis provides insights and methodologies applicable to a wider range of medical imaging challenges, opening avenues for advancements across various diagnostic domains reliant on dynamic tissue movement analysis.

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